Algebra II with Trigonometry Chapter 5.2

Olympiad Problems (GPT-5.4)

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Problems

1. Evaluate exactly:

$$\frac{\sin 15^\circ + \cos 15^\circ}{\sin 15^\circ \cos 15^\circ}.$$

2. A right triangle has acute angle θ with

$$\tan\theta + \cot\theta = 4.$$

Find the exact value of $\sin\theta + \cos\theta$.

3. Two observers stand on the same straight road on opposite sides of a tower. Their distances from the tower differ by 40 m, and the angles of elevation to the top are 45° and 30°. Find the height of the tower.

4. Without a calculator, determine the exact value of

$$(\sec 30^\circ - \tan 30^\circ)(\sec 30^\circ + \tan 30^\circ) + (\csc 45^\circ - \cot 45^\circ)^2.$$

5. In a right triangle, the altitude from the right angle to the hypotenuse divides the hypotenuse into segments of lengths 1 and 4. Find the exact value of the triangle's perimeter.

6. A balloon is vertically above a straight road. The angles of depression to two mileposts on the same side of the balloon are 45° and 30°. How high is the balloon above the road?